

513 Assignment 1

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Deliverable 1

```
Python 3.12.15
Platform Windows-11-10.0.26100-SP0
Executable c:\Users\EMMWILL\repos\EE513\.venv\Scripts\python.exe
```

Figure 1: First Cell

```
[PASS] Python 3.10 or newer found 3.12

True
```

Figure 2: Python Version

```
[PASS] all required packages import numpy 2.5.3, scipy 1.18.1, matplotlib 3.11.2, cv2 5.0.0, rumpy 0.27.1, imageio 2.38.0, skimage 0.26.0
Not used until week 6, but the course environment installs them now: colour 0.4.7
missing: torch, gradio - run 'uv sync' again.
```

Figure 3: Packages

```
[PASS] OpenCV major version is 5 found 5.0.0
[PASS] you have the contrib build, not plain opencv-python cv2.xfeatures2d present
[PASS] functions this course uses are present 7 of 7

cv2.CascadeClassifier present (contrib restores this)
cv2.HOGDescriptor present (contrib restores this)
cv2.AKAZE_create moved cv2.xfeatures2d.AKAZE_create: present
cv2.KAZE_create moved cv2.xfeatures2d.KAZE_create: present
cv2.BRISK_create moved cv2.xfeatures2d.BRISK_create: present

In OpenCV 5 you call cv2.xfeatures2d.AKAZE_create(), not cv2.AKAZE_create().
If a tutorial calls the second form, it predates 2026.
```

Figure 4: OpenCV

Deliverable 2

Link to my repo or <https://github.com/emmwil-star/EE513>

Deliverable 3

```
Opening: C:\Users\EMMWILL\repos\EE513\exercises\01_assignment\images\water2.dng
Raw Type: RawType.Flat
Raw Shape: (3024, 4032)
Raw Dtype: uint16
Raw Pattern: [[2 3]
 [1 0]]
Color Description: b'RGBG'
Black Level Per Channel: [528, 528, 528, 528]
White Level: 4095
```

Figure 5: Deliverable 3

Deliverable 4

Based on the results above, (RawType.Flat, raw_image is 2D, and 2x2 raw_pattern) my image is a Bayer mosaic with a color description of RGBG.

Deliverable 5

Not applicable.

Deliverable 6

	Image .jpeg	Image .dng
Size (MB)	3.47	9.95
ColorBit Depth	uint8	uint12
Number of Channels	3	3

Table 1: Image Metadata

Deliverable 7



Figure 6: Zoomed Blue Water Bottle .jpeg



Figure 7: Zoomed Blue Water Bottle .dng

Figures 6 and 7 show the zoomed portion of the logo of my water bottle. The logo is a slightly darker texture.

Deliverable 8

Three changes that modify the .dng to .jpeg include:

1. Reduction in color bit depth from 12-bit to 8-bit which reduces the number of possible color values.
2. Converting from a raw Bayer mosaic to an RGB image.

Deliverable 9

Deliverable 10

Deliverable 11

Deliverable 12

Deliverable 13